
Modern Machine Learning and Project Directions

A practical map from model families to project design

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Learning objectives

By the end of this lecture, you should be able to:

1. organize common ML methods by data structure and learning setting;
2. recognize the main idea behind several modern model families;
3. choose a realistic model and evaluation strategy for a project;
4. design controlled comparisons that support a clear conclusion.

The goal is to choose methods for a problem, not to memorize model names.

A map of the field

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Problem before model

A machine-learning study starts with four decisions:

1. **Input and output:** what information is available and what should be predicted?
2. **Supervision:** what labels, rewards, or other signals exist?
3. **Evaluation:** what result actually matters?
4. **Resources:** what data and computing are realistically available?

Only then choose the model.

Common mistake

“Use a Transformer” describes a method, not a research question.

Learning settings

Setting	Available signal	Example
supervised	labels for examples	disease classification
weak/semi-supervised	limited or noisy labels	image segmentation
self-supervised	targets constructed from data	masked prediction
unsupervised	no task labels	clustering, density estimation
reinforcement learning	interaction and reward	sequential control

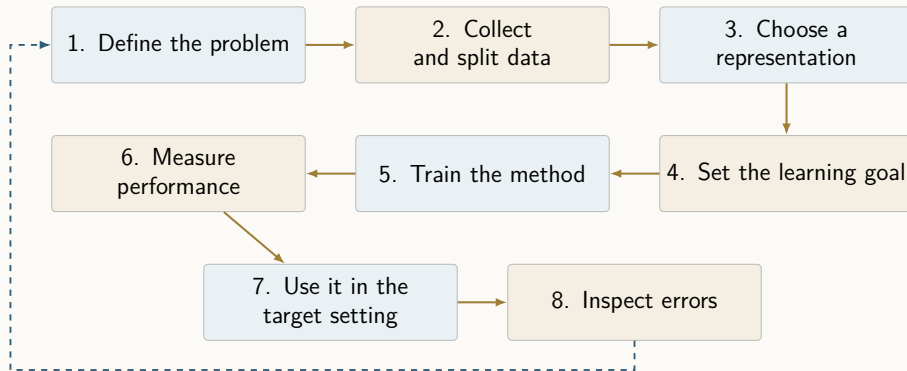
The learning setting is determined by the information available during training.

Data structure guides model choice

Data	Structure to preserve	Common choices
tabular	heterogeneous features	linear, tree, boosting
time series	temporal structure	filters, sequence models
images	spatial locality	CNNs, vision transformers
text	sequence and context	transformers
graphs	relationships	graph neural networks
multiple types	cross-modal alignment	contrastive, fusion models

A model is useful when its structure matches useful structure in the data.

A common machine-learning pipeline



Major model families

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Linear models as strong baselines

Linear models remain useful because they are:

- ▶ fast to train;
- ▶ easy to reproduce;
- ▶ relatively easy to interpret;
- ▶ strong when the representation is informative.

They provide a clear test of whether additional model complexity is actually useful.

How this is used today

A pretrained representation followed by a linear classifier is a common way to test representation quality.

Trees and boosting

A decision tree repeatedly splits the feature space using simple rules such as:

$$x_j < c \quad \text{or} \quad x_j \geq c.$$

- ▶ Trees capture nonlinear feature interactions.
- ▶ Random forests average many trees to reduce instability.
- ▶ Gradient boosting adds trees sequentially to correct current errors.

For tabular data, boosted trees are often an important comparison method.

Neural networks

A neural-network layer applies:

$$\mathbf{h}^{(\ell+1)} = \phi \left(\mathbf{W}^{(\ell)} \mathbf{h}^{(\ell)} + \mathbf{b}^{(\ell)} \right).$$

The nonlinear function ϕ allows multiple layers to represent nonlinear relationships.

Neural networks can learn:

representation + prediction

Greater capacity can model richer patterns, but usually requires more data and stronger regularization.

Convolution and spatial locality

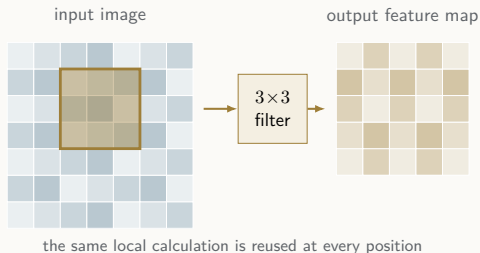
A convolution applies the same local filter at many positions:

$$y[t] = \sum_s w[s]x[t - s].$$

This gives:

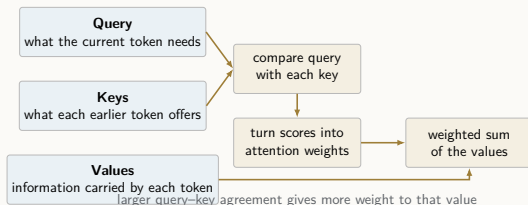
- ▶ local feature extraction;
- ▶ parameter sharing across locations.

Typical uses include signals, images, volumes, and video.



Attention

Attention allows one item to collect information from other items.



The three roles are:

- ▶ **query**: what information is needed;
- ▶ **key**: what each item offers for matching;
- ▶ **value**: the information that may be combined.

Scaled dot-product attention

Collect queries, keys, and values into Q , K , and V :

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d}}\right) V.$$

The computation is:

$$QK^T \rightarrow \text{matching scores} \rightarrow \text{normalized weights} \rightarrow \text{weighted values}.$$

Attention therefore builds a representation by deciding which other items matter most.

Representation learning

A representation is a learned numerical description:

$$z = f_{\theta}(x).$$

Common learning signals include:

- ▶ reconstruction;
- ▶ predicting hidden or future content;
- ▶ contrastive learning;
- ▶ supervised prediction.

The goal is:

preserve information useful for later tasks.

How this is used today

PCA learns a linear representation; neural networks can learn nonlinear representations.

Pretraining and transfer

Pretraining learns reusable parameters from a large source dataset.

A target task can then:

- ▶ freeze the representation and train a new head;
- ▶ fine-tune the full model;
- ▶ update only a small adapter;
- ▶ use prompting when the model supports it.

Foundation models extend this idea by pretraining broadly for reuse across many tasks.

For a course project, adapting an existing model is usually more realistic than training one from scratch.

Multimodal models

A multimodal model combines several data types, for example:

text + images + audio.

The key problem is to learn useful relationships across modalities.

Potential benefits:

- ▶ complementary information;
- ▶ reusable representations;
- ▶ reduced task-specific labeling.

The useful project question is whether the additional modality produces measurable improvement.

Generative models

Generative models describe how observations may be produced.

Common forms include:

- ▶ **autoregressive:** predict the next element from previous elements;
- ▶ **latent-variable:** explain observations using hidden variables;
- ▶ **diffusion:** learn to reverse a gradual noising process.

For example:

$$p(x) = \prod_t p(x_t | x_{<t})$$

is an autoregressive factorization.

GMMs from earlier weeks are already latent-variable generative models.

Choosing a project

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From idea to testable project

Weak:

“Apply deep learning to medical images.”

Stronger:

Does a pretrained representation improve lesion classification over a linear model and a small CNN when only 20% of labels are available?

A project direction should identify:

- ▶ the task;
- ▶ the dataset;
- ▶ the comparison;
- ▶ the performance measure;
- ▶ the main constraint or limitation.

Dataset selection

Before choosing the model, check the data.

- ▶ What is one independent sample?
- ▶ What is the correct train/test split unit?
- ▶ Are labels reliable?
- ▶ Are important groups represented?
- ▶ Can related samples leak across splits?

Common mistake

Frames from the same video or records from the same patient should usually not be split randomly across training and testing.

Performance measures follow the goal

Goal	Useful measures	Common warning
balanced classes	accuracy; class-averaged score	inspect each class
rare events	sensitivity; precision	accuracy can mislead
numerical prediction	absolute or squared error	scale and outliers matter
image regions	overlap and boundary error	empty regions need a rule
search or ranking	top- k relevance	candidate set matters
probabilities	probability error, calibration	averages hide subgroups

A performance measure should reflect the decision the model is intended to support.

Controlled comparison ladder

A useful study compares increasing levels of complexity:

1. trivial reference;
2. simple classical model;
3. strong standard method;
4. proposed model or adaptation.

Keep the data split, preprocessing, evaluation, and training budget as comparable as possible.
Then change one main factor at a time:

one controlled change → one interpretable conclusion

Compute-aware project scope

Estimate cost before implementation:

- ▶ dataset size and preprocessing time;
- ▶ memory requirements;
- ▶ training time per run;
- ▶ number of baselines and repeated runs;
- ▶ inference cost if deployment matters.

When resources are limited, use a smaller model, fixed pretrained features, or a smaller controlled dataset.

A completed controlled experiment is more useful than an unfinished large one.

Error analysis

Average scores do not explain model behavior.

Inspect:

- ▶ class-specific errors;
- ▶ high-confidence mistakes;
- ▶ difficult acquisition conditions;
- ▶ cases where methods disagree;
- ▶ sensitivity to preprocessing.

The purpose is to identify:

where and why the model fails.

Possible project directions

- ▶ **Tabular:** linear models versus boosted trees.
- ▶ **Time series:** anomaly detection under noise or missing values.
- ▶ **Text:** count-based versus pretrained representations.
- ▶ **Images:** transfer learning under limited labels.
- ▶ **Signals:** handcrafted versus learned features.
- ▶ **Scientific data:** robustness across locations, devices, or seasons.
- ▶ **Multimodal:** measure whether an additional modality adds useful information.

The project question should be narrower than the application domain.

Responsible and repeatable work

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Repeatable experiments

Another person should be able to reconstruct the reported result.

Record:

- ▶ dataset version and split;
- ▶ preprocessing;
- ▶ software environment;
- ▶ model configuration;
- ▶ random seeds;
- ▶ hardware;
- ▶ results from relevant runs.

result $\longleftrightarrow$ exact configuration

Ethics and responsible scope

Before using a dataset or model, ask:

- ▶ Do we have permission to use the data?
- ▶ Could labels contain historical or measurement bias?
- ▶ Who is affected by false positives and false negatives?
- ▶ Does the proposed use go beyond the available evidence?
- ▶ Is human review needed?

Common mistake

Publicly downloadable does not automatically mean unrestricted, representative, or ethically appropriate.

Project readiness

Before implementation, the group should be able to state:

1. **Question:** one testable claim.
2. **Data:** sample unit, labels, and split strategy.
3. **Methods:** baseline and main comparison.
4. **Evaluation:** primary measure and error analysis.
5. **Resources:** compute estimate and fallback plan.
6. **Risks:** leakage, ethics, licensing, and limitations.

If one item is unclear, resolve it before increasing model complexity.

Summary

Three principles organize this lecture:

1. **Match structure:** choose models that fit the data and learning signal.
2. **Control comparisons:** change one important factor at a time.
3. **Design for evidence:** evaluation, error analysis, and reproducibility matter as much as the model.

sharp question + controlled experiment > oversized model

A sharp question beats an oversized model

Questions and discussion